
Capstone Project – Pre-Submission Document

BuildMax Infrastructure Ltd. – Predictive Project Delivery Analytics

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1. Business Problem Understanding and Objective Formulation

1.1 Domain & Business Context

BuildMax Infrastructure Ltd. is a large-scale EPC (Engineering, Procurement and Construction) organisation delivering commercial buildings, transportation and airport infrastructure, industrial facilities, residential developments, and public-sector (PSU) projects across six business areas. The organisation works end-to-end — from Concept to Commissioning — typically on fixed-price, fixed-timeline contracts with strict quality commitments.

Rapid growth has intensified project-execution complexity, and executive leadership has flagged a systemic, company-wide rise in schedule delays and cost overruns. Project delivery in some cases is exceeding planned timelines by months, and the resulting over-budget execution has significantly eroded profitability. Monitoring today is largely periodic and manual, so critical risks surface only after significant financial and reputational damage has already occurred. Historical data captured across project management, procurement, finance, contractor management, safety, and site-operations systems presents an opportunity to build a predictive, data-driven risk-management framework.

1.2 Identified Business Problems

- Frequent project schedule overruns, in some cases extending by months beyond planned completion.
- Persistent cost overruns / over-budget delivery, eroding margins and triggering contractual disputes and reputational damage.
- Reactive, periodic monitoring with no early-warning signals — risk compounds undetected between inspections.
- Siloed data across project management, procurement, finance, safety, and site-operations systems.
- Inconsistent contractor and vendor performance contributing to procurement delays and productivity shortfalls.
- Material procurement delays and fluctuating labour productivity across project sites.
- Safety incidents causing project disruptions and downstream schedule impact.
- No data-driven mechanism currently exists to flag high-risk projects before critical thresholds are breached.

1.3 Formulated Objectives

Business Objectives:

- Predict which active projects are at risk of schedule delay or cost overrun before critical thresholds are breached.
- Reduce overall project delays and budget overruns to protect margins, reputation, and customer commitments.
- Identify the top controllable drivers of delay and cost escalation to enable targeted interventions.
- Establish a data-driven early-warning system integrated into project-management workflows.
- Improve resource utilisation, contractor selection, and decision-making through data-driven insights.

Analytical Objectives:

- Build a binary classification model to flag projects at high risk of schedule delay (target: delayed vs. on-time).
- Develop a regression model to quantify the expected cost-overrun percentage for flagged projects.
- Apply SHAP-based interpretability to rank and explain feature contributions at the project level.
- Perform unsupervised clustering to segment projects by risk profile for portfolio-level monitoring.

- Identify the most influential variables affecting project delay and cost outcomes for business reporting.

2. Challenges and Constraints

2.1 Data Challenges

- Missing or inconsistent records across heterogeneous source systems (project management, ERP, procurement, site logs).
- Duplicate records inflating the raw dataset and requiring de-duplication before modelling.
- Significant blank/missing values in key fields such as `supplier_performance_score`, requiring imputation (mean/average substitution).
- Several administrative / legacy identifier fields — `legacy_project_code`, `reporting_batch_id`, `regional_cluster_id`, and `administrative_reference` — are shared indiscriminately across project IDs irrespective of project type or region, indicating unreliable provenance and a risk of noise or multicollinearity.
- Class imbalance in the delay label, since on-time projects are typically far more frequent than delayed ones.
- Temporal leakage risk: features captured after a delay event must be excluded from training to prevent data leakage.

Data Profiling Snapshot (initial analysis):

Metric	Observation
Total records received	50,500
Duplicate rows identified	500
Usable records after de-duplication	50,000
<code>supplier_performance_score</code>	~5,011 blank values — to be imputed using mean / average
Shared-identifier issue	<code>legacy_project_code</code> , <code>reporting_batch_id</code> , <code>regional_cluster_id</code> , and <code>administrative_reference</code> are shared across project IDs irrespective of project type / region

2.2 Modelling & Implementation Constraints

- Ground-truth labels (delay, overrun) require careful definition — absolute vs. relative thresholds must be agreed with stakeholders.
- Short project lifecycles may limit the look-back window available for lag-based features.
- High-dimensional dataset with many variables of varying relevance, requiring feature-selection techniques (e.g., PCA) to control noise.
- Model interpretability is paramount — black-box predictions will face low adoption by project managers; explainability is a hard requirement.
- Likely resistance to adopting analytics-driven decisions within a traditionally manual, experience-led organisation.
- Dynamic project environments affected by external factors (weather, regulatory, market conditions) that are difficult to fully capture.
- Deployment requires integration with live data pipelines that may not currently exist.

2.3 Assumptions & Limitations

- Historical project records are representative of future project profiles; no major structural business-model shifts are assumed.

- A project is defined as delayed if it exceeds planned duration by more than 10%; the cost-overflow threshold is set at 5% above budget.
- Data from safety, procurement, and finance systems can be linked via a common project identifier.
- Data calculations already present in the source sheets (e.g., percentage fields) are assumed accurate, and dependencies between columns can be derived directly from the data.
- Relevant risk factors are assumed to be captured within the available dataset; external or unforeseen events (policy changes, natural disasters) may not be fully predictable and fall outside the model's scope.
- Analysis and conclusions are limited to the provided dataset; prediction accuracy depends on overall data quality and completeness.

3. Background Study / Literature Review

Research in construction and EPC project management consistently identifies labour shortages, procurement delays, contractor performance, financial constraints, safety incidents, and environmental disruption as dominant drivers of schedule delay and cost overrun. Traditional techniques such as the Critical Path Method (CPM) and Earned Value Management (EVM) support reactive monitoring but offer limited predictive capability, motivating a shift toward machine-learning-based early-warning approaches.

Key techniques and best practices identified:

- Earned Value metrics (SPI, CPI, EAC) as core engineered features.
- Ensemble methods — Random Forest, XGBoost, LightGBM, Gradient Boosting — as primary classifiers / regressors, alongside baseline Logistic Regression and Decision Trees.
- SHAP values for model explainability and practitioner trust.
- K-Means / DBSCAN clustering for project risk segmentation.
- Rolling-window lag features to capture temporal dynamics within a project lifecycle.
- Artificial Neural Networks (ANN) and SVM explored in industry studies as alternative classification techniques.

Reference papers consulted:

- Breiman, L. — Random Forests — <https://www.stat.berkeley.edu/~breiman/randomforest2001.pdf>
- Chen & Guestrin — XGBoost: A Scalable Tree Boosting System — <https://arxiv.org/pdf/1603.02754>
- Predicting Delays and Cost Overruns in Construction Projects: A Machine Learning Approach — ResearchGate
- International Journal of Engineering Science & Humanities — ijesh.com/ij/article/view/759/800
- International Journal of Scientific Development and Research — [IJEDR2408020](http://www.ijedr.com/papers/IJEDR2408020)
- EKB Journals — [article_433276_a18cb8303f65ae2dcb93399ecef675b](https://www.ekbjournals.com/article_433276_a18cb8303f65ae2dcb93399ecef675b)

4. Proposed Methodology / Solution Approach

4.1 Solution Workflow

The end-to-end analytics pipeline follows seven sequential phases, taking the project from raw multi-source data through to a business-ready early-warning dashboard:

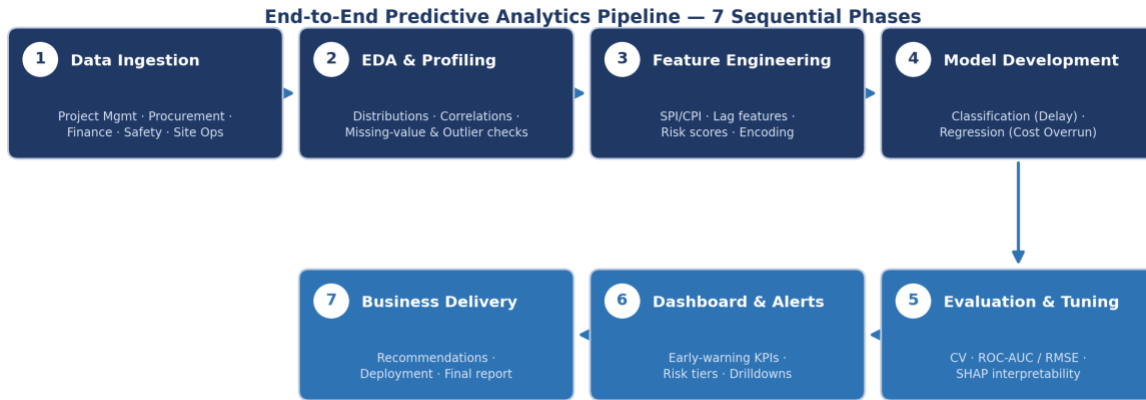


Figure 4.1 — Block diagram of the end-to-end Solution Workflow

1. Data Ingestion	Project Mgmt · Procurement · Finance · Safety · Site Ops records
2. EDA & Profiling	Distribution analysis · Correlation matrix · Missing-value heatmap · Outlier detection
3. Feature Engineering	SPI / CPI derivation · Lag features · Risk composite scores · Categorical encoding
4. Model Development	Classification: Delay Risk (RF / XGBoost) Regression: Cost Overrun %
5. Evaluation & Tuning	CV · ROC-AUC / RMSE · SHAP interpretability · Threshold setting
6. Dashboard & Alerts	Early-warning KPIs · Project risk tier · Drilldown analytics
7. Business Delivery	Recommendations · Deployment proposal · Final report & presentation

4.2 Analytical Approach

- **Task A** – Supervised Classification (Random Forest, XGBoost): Predict binary delay-risk label; optimise for recall to minimise missed high-risk projects.
- **Task B** – Supervised Regression (XGBoost Regressor, Ridge): Predict cost-overrun percentage for projects identified as at-risk.
- **Task C** – Unsupervised Clustering (K-Means): Segment projects into Low / Medium / High risk tiers for portfolio dashboards.
- **Task D** – Interpretability (SHAP): Produce global feature-importance rankings and local per-project explanations for project managers.

Feature engineering will derive EVM metrics (SPI, CPI, EAC), contractor performance scores, procurement delay indices, safety incident rate per labour-hour, weather-impact scores, and rolling 4-week productivity trends. Target leakage will be controlled through strict temporal train/validation splits aligned to project-phase milestones.

5. Proposed Timeline

Week	Activity
1	Business Understanding & Data Review
2	Data Cleaning, De-duplication & Preprocessing

3	Exploratory Data Analysis (EDA) & Profiling
4	Feature Engineering
5	Model Development (Classification & Regression)
6	Model Evaluation, Tuning & SHAP Interpretability
7	Dashboard Development & Risk Tiering
8	Final Report, Recommendations & Presentation

Total Duration: 8 Weeks

6. Expected Deliverables / Outputs

Deliverable	Description
Data Understanding & EDA Report	Data quality assessment, exploratory analysis, and key insights from the BuildMax dataset.
ML Prediction Models (Classification & Regression)	Random Forest and XGBoost based delay-classification and cost-overrun regression models, compared on Accuracy / Precision / Recall / F1 / ROC-AUC / RMSE / R ² to select the best-fit model.
Feature Importance & SHAP Analysis	Global and local explainability highlighting the key drivers of project delay and cost overrun.
Risk Monitoring Dashboard	Interactive visualization of project risk scores, delay probabilities, risk tiers, and key drivers.
Business Recommendations	Actionable strategies to mitigate delay risk and improve project execution and profitability.
Technical Documentation	Methodology, data preparation, model details, and implementation guidelines for reproducibility.
Deployment Recommendation Framework	Proposed approach for integrating the predictive models and dashboard into BuildMax's existing systems for continuous, proactive risk management.
Final Presentation	Consolidated findings, model performance, business insights, and recommendations for stakeholders and evaluators.

Expected Outcome: The project will deliver a predictive analytics framework that helps BuildMax Infrastructure Ltd. proactively identify high-risk projects, reduce delays, control costs, and improve overall project performance and profitability.